Testing and Validation

Testing Approach

The chatbot system was evaluated using a structured testing framework designed to measure performance across key criteria, including routing accuracy, response quality, URL relevance, and suggestion relevance.

A set of predefined test queries representing real-world citizen interactions was created based on common council service use cases such as Council Tax, Waste & Bins, Benefits & Support, Education, and Planning. These queries were used to simulate realistic user behaviour and validate the system's ability to correctly interpret and respond to user inputs.

Testing was conducted iteratively, allowing improvements to be made between runs. Early testing highlighted weaknesses in routing and response accuracy, which were addressed through enhancements to service detection logic, retrieval mechanisms, and response generation.

Initial Testing Results

The initial testing phase consisted of **45 test cases**, producing the following results:

- Routing accuracy: **91.11%**
- Response quality accuracy: **82.22%**
- URL relevance accuracy: **93.33%**
- Suggestion relevance accuracy: **80.00%**
- Final accuracy: **77.78%**

These results indicated strong routing and URL performance, but highlighted limitations in response quality and overall accuracy. As a result, the system did not meet the target benchmark of **85% accuracy** at this stage.

Further analysis showed that certain service areas, particularly **Benefits & Support** and **Education**, had lower response performance, indicating the need for improved retrieval and answer generation in these domains.

System Improvements

Following the initial evaluation, several improvements were implemented:

- Enhanced service routing using stronger keyword triggers
- Improved response generation using LangChain integration
- Refinement of retrieval mechanisms for more relevant context selection
- Addition of guardrails and better handling of ambiguous queries
- Expansion of test coverage to include more scenarios

These changes were aimed at improving both the accuracy and consistency of responses across all service areas.

Final Testing Results

The final testing phase consisted of **160 test cases**, providing a more comprehensive evaluation of system performance.

- Total tests: **160**
- Passed: **136**
- Overall accuracy: **85.00%**
- Target met (85%)